# Supplementary material for ”The Relationship Between Intergenerational Mobility and Equality of Opportunity”, published in *The Scandinavian Journal of Economics*

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Table S.1: Summary statistics, sibling sample

| | Mean | Std. dev. | Min | Max |
|---|---|---|---|---|
| **Panel A. Child** | | | | |
| Birth year | 1972.6 | 4.2 | 1965.0 | 1980.0 |
| Income | 199 | 123 | 94 | 37,230 |
| Share women | 48% | | | |
| **Panel B. Mother** | | | | |
| Birth year | 1946.3 | 4.9 | 1925.0 | 1962.0 |
| Income | 217 | 70 | 94 | 2,040 |
| Years of schooling | 11.1 | 2.7 | 7.0 | 20.0 |
| Age at birth | 26.2 | 4.3 | 18.0 | 40.0 |
| **Panel C. Father** | | | | |
| Birth year | 1943.9 | 5.2 | 1925.0 | 1962.0 |
| Income | 322 | 151 | 97 | 20,642 |
| Years of schooling | 10.8 | 3.0 | 7.0 | 20.0 |
| Age at birth | 28.6 | 4.5 | 18.0 | 40.0 |
| **Panel D. Family** | | | | |
| Family size | 2.3 | 0.5 | 2.0 | 10.0 |
| Parents divorced | 22% | | | |
| Parent died | 12% | | | |
| Same parish | 88% | | | |

*Note:* Table shows summary statistics for the individual-level data in the siblings sample.

Table S.2: IOp estimates

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| **Panel A. National IOp** | | | | | | | | | |
| *Gini* | | | | | | | | | |
| Absolute IOp | 0.069 | 0.074 | 0.075 | 0.077 | 0.080 | 0.071 | 0.076 | 0.083 | 0.075 |
| Relative IOp | 0.353 | 0.376 | 0.383 | 0.393 | 0.409 | 0.362 | 0.385 | 0.421 | 0.382 |
| Inequality | 0.197 | 0.197 | 0.197 | 0.197 | 0.197 | 0.197 | 0.196 | 0.196 | 0.196 |
| *Mean log deviation* | | | | | | | | | |
| Absolute IOp | 0.008 | 0.009 | 0.009 | 0.009 | 0.010 | 0.008 | 0.009 | 0.011 | 0.009 |
| Relative IOp | 0.129 | 0.136 | 0.144 | 0.147 | 0.157 | 0.120 | 0.143 | 0.166 | 0.142 |
| Inequality | 0.065 | 0.065 | 0.065 | 0.065 | 0.065 | 0.065 | 0.065 | 0.065 | 0.065 |
| **Panel B. Mean IOp** | | | | | | | | | |
| *Gini* | | | | | | | | | |
| Absolute IOp | 0.048 | 0.050 | 0.050 | 0.042 | 0.043 | 0.036 | 0.036 | 0.042 | 0.043 |
| Relative IOp | 0.264 | 0.275 | 0.273 | 0.227 | 0.236 | 0.196 | 0.195 | 0.231 | 0.236 |
| Inequality | 0.182 | 0.182 | 0.182 | 0.182 | 0.182 | 0.182 | 0.181 | 0.181 | 0.181 |
| *Mean log deviation* | | | | | | | | | |
| Absolute IOp | 0.004 | 0.004 | 0.004 | 0.003 | 0.003 | 0.002 | 0.002 | 0.003 | 0.003 |
| Relative IOp | 0.078 | 0.079 | 0.077 | 0.055 | 0.058 | 0.042 | 0.044 | 0.057 | 0.060 |
| Inequality | 0.054 | 0.054 | 0.054 | 0.054 | 0.054 | 0.054 | 0.054 | 0.054 | 0.054 |
| **Circumstances** | | | | | | | | | |
| Parental income | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ |
| Parental education | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Parental occupation | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Family characteristics | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Gender | | | | | ✓ | | | | |
| Skills | | | | | | | | ✓ | |
| Parental capital income | | | | | | | | | ✓ |
| No. of observations | 1,077,046 | 1,077,046 | 1,077,046 | 1,077,046 | 1,077,046 | 1,077,046 | 501,591 | 501,591 | 947,192 |

*Note:* The table shows IOp and inequality estimates, using the *Gini* or *mean logarithmic deviation* as the inequality index. Panel A. shows estimates at the national level, while Panel B. shows averages across local labor markets. *Parental income*, *education*, *occupation*, and *capital income* includes the variable for both parents separately. *Family characteristics* includes family size, both parents' year of birth and age when the child was born, and indicators for early parental death, divorce during childhood, and living in the same parish as both parents during childhood.

Table S.3: Circumstances

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
|---|---|---|---|---|---|---|---|---|---|
| **Panel A. Absolute IOp** | | | | | | | | | |
| IGE | 0.86 | 0.84 | 0.84 | 0.84 | 0.83 | 0.80 | 0.73 | 0.72 | 0.82 |
| | (0.05) | (0.05) | (0.05) | (0.05) | (0.05) | (0.05) | (0.06) | (0.06) | (0.05) |
| Rank persistence | 0.86 | 0.86 | 0.86 | 0.88 | 0.87 | 0.87 | 0.81 | 0.80 | 0.85 |
| | (0.05) | (0.05) | (0.05) | (0.04) | (0.04) | (0.04) | (0.05) | (0.05) | (0.05) |
| Sibling correlation | 0.77 | 0.79 | 0.79 | 0.81 | 0.81 | 0.79 | 0.68 | 0.69 | 0.80 |
| | (0.06) | (0.06) | (0.05) | (0.05) | (0.05) | (0.05) | (0.07) | (0.07) | (0.05) |
| **Panel B. Relative IOp** | | | | | | | | | |
| IGE | 0.89 | 0.84 | 0.84 | 0.82 | 0.81 | 0.76 | 0.73 | 0.70 | 0.82 |
| | (0.04) | (0.05) | (0.05) | (0.05) | (0.05) | (0.06) | (0.06) | (0.06) | (0.05) |
| Rank persistence | 0.90 | 0.89 | 0.89 | 0.88 | 0.87 | 0.85 | 0.81 | 0.80 | 0.87 |
| | (0.04) | (0.04) | (0.04) | (0.04) | (0.04) | (0.05) | (0.05) | (0.05) | (0.04) |
| Sibling correlation | 0.73 | 0.74 | 0.76 | 0.77 | 0.76 | 0.73 | 0.62 | 0.62 | 0.77 |
| | (0.06) | (0.06) | (0.06) | (0.06) | (0.06) | (0.06) | (0.07) | (0.07) | (0.06) |
| **Circumstances** | | | | | | | | | |
| Parental income | ✓ | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ |
| Parental education | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Parental occupation | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Family characteristics | | | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Gender | | | | | ✓ | | | | |
| Skills | | | | | | | | ✓ | |
| Parental capital income | | | | | | | | | ✓ |
| No. of observations | 1,077,046 | 1,077,046 | 1,077,046 | 1,077,046 | 1,077,046 | 1,077,046 | 501,591 | 501,591 | 947,192 |

*Note:* *Parental income*, *education*, *occupation*, and *capital income* includes the variable for both parents separately. *Family characteristics* includes family size, both parents' year of birth and age when the child was born, and indicators for early parental death, divorce during childhood, and living in the same parish as both parents during childhood.

Table S.4: By sample size

| | Absolute IOp | | | Relative IOp | | |
|---|---|---|---|---|---|---|
| | IGE | Rank | Sibl | IGE | Rank | Sibl |
| **Panel A. N $\leq$ 3,396** | | | | | | |
| Unweighted | 0.41 | 0.44 | 0.10 | 0.34 | 0.46 | 0.05 |
| | (0.12) | (0.12) | (0.12) | (0.12) | (0.11) | (0.12) |
| Weighted | 0.56 | 0.53 | 0.22 | 0.51 | 0.55 | 0.19 |
| | (0.11) | (0.11) | (0.11) | (0.11) | (0.11) | (0.11) |
| **Panel B. N $>$ 3,396** | | | | | | |
| Unweighted | 0.71 | 0.77 | 0.59 | 0.70 | 0.79 | 0.50 |
| | (0.09) | (0.08) | (0.12) | (0.09) | (0.08) | (0.12) |
| Weighted | 0.86 | 0.88 | 0.85 | 0.85 | 0.89 | 0.83 |
| | (0.07) | (0.06) | (0.08) | (0.07) | (0.06) | (0.08) |

*Note:* Each cell shows the correlation across local labor markets (LLM) between the measures of IOp and mobility indicated in the header, with standard errors in parentheses. "Weighted" rows show correlations weighted by the no. of observations of the LLM. The sample has been split at the median LLM size in two parts, with 63 LLMs each. The smaller LLMs are shown in Panel A., and the larger in Panel B. The IOp, IGE, and rank estimators use a different sample than the sibling correlation: no. of observations is 99,826 and 109,031 respectively in Panel A.; and 977,220 and 657,974 in Panel B.

Table S.5: Bootstrap simulation, 100 obs./LLM

| | Absolute IOp | | Relative IOp | |
|---|---|---|---|---|
| | IGE | Rank | IGE | Rank |
| A. Unweighted | 0.54 | 0.42 | 0.52 | 0.45 |
| | (0.08) | (0.09) | (0.07) | (0.07) |
| B. Weighted | 0.56 | 0.45 | 0.53 | 0.45 |
| | (0.14) | (0.17) | (0.14) | (0.16) |
| C. N $\leq$ 3,396 | 0.52 | 0.40 | 0.49 | 0.42 |
| | (0.11) | (0.11) | (0.10) | (0.10) |
| D. N $>$ 3,396 | 0.56 | 0.45 | 0.53 | 0.47 |
| | (0.10) | (0.10) | (0.10) | (0.09) |

*Note:* The table shows versions of the main results where we enforce a uniform sample size of 100 observations from each local labor market by sampling with replacement. The procedure is repeated 500 times, and each cell shows mean correlations and standard errors from the bootstrap distributions. Panel A. shows unweighted correlations, while Panel B. shows correlations weighted using sample sizes from the full data. Panels C. and D. show correlations separately for smaller and larger local labor markets (in the original data), as in Table S.4.

Table S.6: Nonlinearities

| | Bottom 25 | | Top 25 | |
|---|---|---|---|---|
| | Absolute IOp | Relative IOp | Absolute IOp | Relative IOp |
| IGE | 12 | 12 | 12 | 12 |
| Rank | 14 | 15 | 13 | 14 |
| Sibl | 10 | 9 | 9 | 10 |

*Note:* Each cell shows the number of regions that are among the bottom (top) 25 regions in terms of both IOp and mobility for the indicated pairs of measures.

Table S.7: Correlations by gender, balanced sample

| | Absolute IOp | | | Relative IOp | | |
|---|---|---|---|---|---|---|
| | IGE | Rank | Sibl | IGE | Rank | Sibl |
| **Panel A. Men** | | | | | | |
| Unweighted | 0.46 | 0.48 | 0.10 | 0.44 | 0.47 | 0.07 |
| | (0.08) | (0.08) | (0.09) | (0.08) | (0.08) | (0.09) |
| Weighted | 0.67 | 0.75 | 0.71 | 0.68 | 0.75 | 0.65 |
| | (0.07) | (0.06) | (0.06) | (0.07) | (0.06) | (0.07) |
| **Panel B. Women** | | | | | | |
| Unweighted | 0.36 | 0.38 | 0.16 | 0.34 | 0.37 | 0.14 |
| | (0.08) | (0.08) | (0.09) | (0.08) | (0.08) | (0.09) |
| Weighted | 0.66 | 0.70 | 0.49 | 0.63 | 0.68 | 0.46 |
| | (0.07) | (0.06) | (0.08) | (0.07) | (0.07) | (0.08) |

*Note:* Each cell shows the correlation across 126 local labor markets between the measures of IOp and mobility indicated in the header, with standard errors in parentheses. ”Weighted” rows show correlations weighted by the no. of observations of the local labor market. The same sample is used for measure. No. of observations is 212,406 in Panel A.; and 183,809 in Panel B.

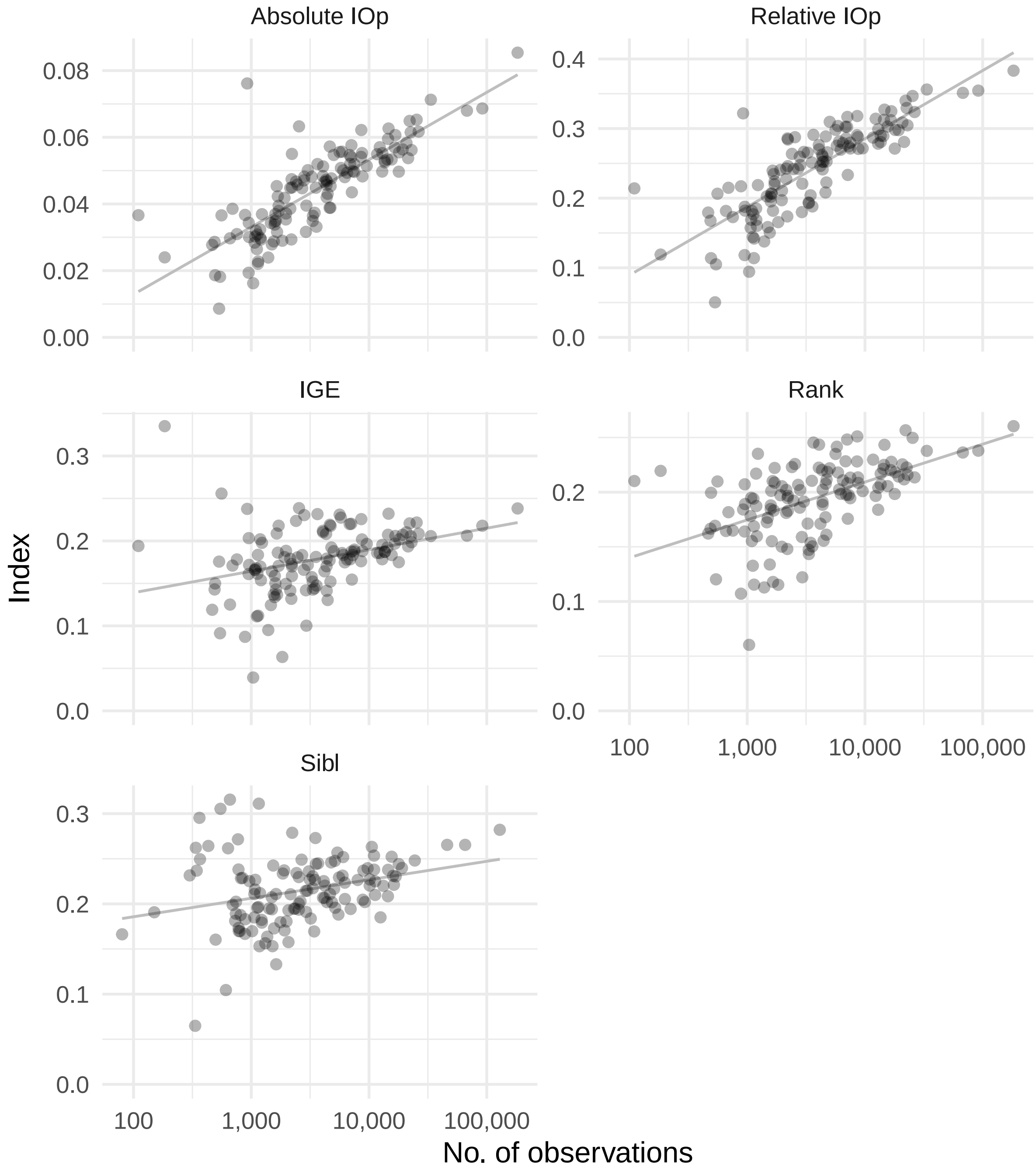


Figure S.1: Relationship between IOp and mobility indices and region size

*Notes:* Each circle represents a local labor market. Lines show OLS regressions of the indicated persistence or IOp measures on sample size.